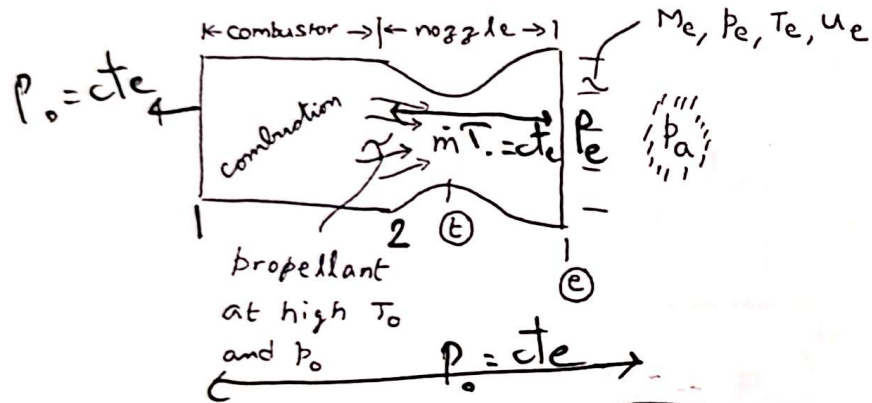


Chemical Rocket Thrust Chamber (ch 11)

* Consists of the combustion chamber and nozzle.



$$u_e = \sqrt{2c_p \left(T_{01} + \frac{Q_R}{c_p} \right) \left[1 - \left(\frac{p_e}{p_{02}} \right)^{(\gamma-1)/\gamma} \right]} \approx \sqrt{2Q_R \left[1 - \left(\frac{p_e}{p_{02}} \right)^{(\gamma-1)/\gamma} \right]} \quad (11.4)$$

$$T_{01} \ll Q_R/c_p$$

$$u_e \approx \sqrt{2 \frac{Q_R}{M} \left[1 - \left(\frac{p_e}{p_{02}} \right)^{(\gamma-1)/\gamma} \right]} \quad (11.5)$$

$$\dot{m} = \frac{A^* p_{02}}{\sqrt{RT_{02}}} \sqrt{\gamma \left(\frac{2}{\gamma+1} \right)^{(\gamma+1)/(\gamma-1)}} \quad (11.6)$$

$$\mathcal{T} = \dot{m} u_e + (p_e - p_a) A_e$$

Eqs. (11.3) and (11.6)

Verify 11.7

(I)

$$\tau = \frac{\mathcal{T}}{A^* p_0} = \sqrt{\frac{2\gamma^2}{\gamma-1} \left(\frac{2}{\gamma+1} \right)^{(\gamma+1)/(\gamma-1)}} \left[1 - \left(\frac{p_e}{p_0} \right)^{(\gamma-1)/\gamma} \right] + \left(\frac{p_e}{p_0} - \frac{p_a}{p_0} \right) \frac{A_e}{A^*} \quad (11.7)$$

$$\dot{Q} = \dot{m}(h_{02} - h_{01}) = \dot{m}c_p(T_{02} - T_{01})$$

$$T_{02} = T_{01} + \frac{Q_R}{c_p}$$

$$h_{02} = h_{0e} \quad \text{or} \quad \frac{u_e^2}{2} = h_{02} - h_e = c_p(T_{02} - T_e) = c_p T_{02} \left(1 - \frac{T_e}{T_{02}} \right)$$

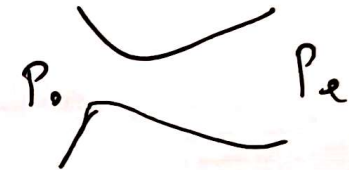
$$u_e = \sqrt{2c_p T_{02} \left[1 - \left(\frac{p_e}{p_{02}} \right)^{(\gamma-1)/\gamma} \right]} = c_p T_{02} \left(1 - \left(\frac{p_e}{p_{02}} \right)^{(\gamma-1)/\gamma} \right)^{1/2}$$

$$u_e = \sqrt{\frac{2\gamma R}{(\gamma-1)M}} T_{02} \left[1 - \left(\frac{p_e}{p_{02}} \right)^{(\gamma-1)/\gamma} \right] \quad (11.3)$$

$$U_e = \sqrt{2C_p T_{02} \left[1 - \left(\frac{p_e}{p_{02}} \right)^{\frac{\gamma-1}{\gamma}} \right]}$$

$$U_e = \sqrt{\frac{2\gamma\bar{R}}{(\gamma-1)M} T_{02} \left[1 - \left(\frac{p_e}{p_{02}} \right)^{\frac{\gamma-1}{\gamma}} \right]}$$

$$U_e \approx \sqrt{2Q_R \left[1 - \left(\frac{p_e}{p_{02}} \right)^{\frac{\gamma-1}{\gamma}} \right]} = \sqrt{2 \frac{\bar{Q}_R}{M} \left[1 - \left(\frac{p_e}{p_{02}} \right)^{\frac{\gamma-1}{\gamma}} \right]}$$



$$\frac{p_e}{p_{02}} \downarrow$$

- For high U_e (high I_{sp}), desire
 - Propellants with low molecular weight, M
 - Propellant mixtures with large heat release, Q_R
 - High combustion chamber pressure, P_{02}

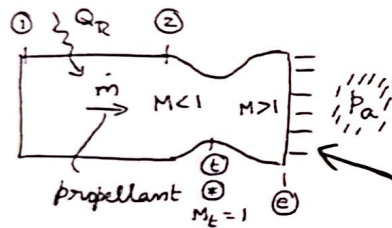
NOTE: Sometimes subscript 2 is dropped, but still conditions in combustion chamber

Combustion Chamber

Characteristic velocity $c^* \stackrel{\text{def}}{=} \frac{p_0 A_t}{\dot{m}}$

Thrust coefficient $C_T \stackrel{\text{def}}{=} \frac{T}{p_0 A_t}$

$$T = C_T p_0 A_t = \dot{m} c^* C_T, \quad T = \dot{m} c^* C_T$$



Q_R - heat added per unit propellant mass

I law across combustor:

$$\dot{m} c_p T_{01} + \dot{m} Q_R = \dot{m} c_p T_{02}$$

$$\Rightarrow T_{02} = T_{01} + \frac{Q_R}{c_p} \quad (11.1)$$

$$p_{02} = p_{01}$$

T_{02} and p_{02} are referred to as chamber conditions, and are often represented as just T_0 and p_0 . T_{02} is also known as the flame temperature.

* Model (ideal rocket thrust chamber)

① Working fluid (propellant) is a perfect gas with constant composition.

② Combustion is equivalent to ideal, constant-pressure heat addition. p_0 constant ($M=0$)

③ Nozzle expansion is steady, one-dimensional and isentropic.

recall Rayleigh-line result (from AE 308)

The Mach number of the flow at the nozzle exit is found from either of the following stipulations:

① If nozzle area ratio is specified, this will determine M_e :

$$\frac{A_e}{A_t} = \frac{A_e}{A^*} = \frac{1}{M_e} \left[\frac{2}{\gamma+1} \left(1 + \frac{\gamma-1}{2} M_e^2 \right) \right]^{\frac{\gamma+1}{2(\gamma-1)}}$$

$$M_e > 1$$

② If the jet pressure at nozzle exit is known, this will determine M_e .

$$\frac{P_o}{P_e} = \frac{P_{oe}}{P_e} = \left(1 + \frac{\gamma-1}{2} M_e^2\right)^{\frac{\gamma}{\gamma-1}}$$

For isentropic nozzle (third assumption in the model),

$$T_{oe} = T_{o2} \text{ and } P_{oe} = P_{o2}.$$

I law across nozzle: $h_{oe} = h_{o2} + \cancel{\frac{u_e^2}{2}}$

$$(or) h_e + \frac{u_e^2}{2} = h_{o2} \Rightarrow c_p T_e + \frac{u_e^2}{2} = c_p T_{o2}$$

$$(or) u_e = \sqrt{2 c_p (T_{o2} - T_e)} \quad \left\{ T_e = \frac{T_{o2}}{1 + \frac{\gamma-1}{2} M_e^2} \right\}$$

Comments

$$① c_p = \frac{\gamma R}{\gamma-1} = \frac{\gamma}{\gamma-1} \frac{\bar{R}}{\bar{M}} \Rightarrow u_e = \sqrt{\left(\frac{2\gamma}{\gamma-1} \frac{\bar{R}}{\bar{M}}\right) (T_{o2} - T_e)}$$

Low molar mass ($\bar{M}$) yields higher u_e .

$$② T_{o2} = T_{o1} + \frac{\bar{Q}_R}{c_p} = T_{o1} + \frac{1}{c_p} \frac{\bar{Q}_R}{\bar{M}}$$

Higher $\left(\frac{\bar{Q}_R}{\bar{M}}\right)$ yields higher T_{o2} , and, hence, higher u_e .

Both $\bar{Q}_R$ and $\bar{M}$ depend on fuel-oxidizer ratio of the reactant mixture.

$$C^* \rightarrow C.C$$

$$C_c \rightarrow N.$$

$$\dot{m} = \frac{A^* p_{0c}}{\sqrt{RT_{0c}}} \sqrt{\gamma \left(\frac{2}{\gamma+1} \right)^{(\gamma+1)/(\gamma-1)}}, \quad (11.6)$$

characteristic velocity $\underline{c^*} \stackrel{\text{def}}{=} \frac{p_0 A_t}{\dot{m}}$

For the ideal rocket thrust chamber model,

2nd
Verify

$$c^* = \sqrt{\frac{1}{\gamma} \left(\frac{\gamma+1}{2} \right)^{\frac{\gamma+1}{\gamma-1}} \frac{\bar{R} T_{02}}{M}} \quad (11.9)$$

{ recall T_{02} is the same as T_0 }

c^* is primarily dependent on combustion chamber properties (temperature, molar mass), and, therefore, it is a performance measure of the combustor part of the thrust chamber.

Thrust coefficient $C_T \stackrel{\text{def}}{=} \frac{T}{p_0 A_t}$

$$\frac{T}{A^* p_0} = \sqrt{\frac{2\gamma^2}{\gamma-1} \left(\frac{2}{\gamma+1}\right)^{\frac{(\gamma+1)(\gamma-1)}{\gamma}} \left[1 - \left(\frac{p_e}{p_0}\right)^{\frac{(\gamma-1)}{\gamma}}\right]} + \left(\frac{p_e}{p_0} - \frac{p_a}{p_0}\right) \frac{A_e}{A^*} \quad (11.7)$$

For the ideal rocket thrust chamber model,

$$C_T = \sqrt{\frac{2\gamma^2}{\gamma-1} \left(\frac{2}{\gamma+1}\right)^{\frac{\gamma+1}{\gamma-1}} \left[1 - \left(\frac{p_e}{p_{02}}\right)^{\frac{\gamma-1}{\gamma}}\right]} + \frac{\frac{p_a}{p_{02}} (p_e - p_a) \frac{A_e}{A^*}}{p_{02}} \quad (11.11)$$

{recall p_{02} is the same as p_0 }

C_T is dependent on nozzle characteristics and so, it is a measure of the performance of the nozzle part of the thrust chamber.

$$T = \dot{m} u_e + \underline{(p_e - p_a) A_e}$$

$$\frac{p_e}{p_0} = \left(1 + \frac{\gamma-1}{2} M_e^2\right)^{\frac{\gamma-1}{\gamma}} \quad \left. \begin{array}{l} \text{put } M_e = 1 \\ \text{for convergent nozzle} \end{array} \right\} \quad \frac{p_e}{p_0} = \left(\frac{\gamma+1}{2}\right)^{\frac{\gamma-1}{\gamma}} = \left(\frac{2}{\gamma+1}\right)^{\frac{\gamma}{\gamma-1}}$$

for convergent nozzle

If the nozzle does not have a diverging portion, then the pressure ratio p_e/p_0 is given by Eq. (3.11) with $M = 1$:

$$p_e/p_0 = \left(\frac{2}{\gamma+1}\right)^{\gamma/(\gamma-1)}$$

and the thrust coefficient reduces to

$$C_{F_{\text{conv}}} = (\gamma+1) \left(\frac{2}{\gamma+1}\right)^{\gamma/(\gamma-1)} - \frac{p_a}{p_0}$$

Verify 11.12

$$\frac{\tau}{P_0 A^*} = C_F = \sqrt{\frac{2\gamma^2}{\gamma-1} \left(\frac{2}{\gamma+1}\right)^{\frac{\gamma+1}{\gamma-1}} \left[1 - \left(\frac{p_e}{p_{02}}\right)^{\frac{\gamma-1}{\gamma}}\right]} + \frac{p_a}{p_{02}} \frac{A_e}{A^*} \quad (11.11)$$

in which $C_{F_{\text{conv}}}$ is the thrust coefficient for a converging nozzle. The ratio of thrusts, with and without a diverging section, is given by

$$\frac{\tau}{\tau_{\text{conv}}} = \frac{C_F}{C_{F_{\text{conv}}}} = \frac{11.11}{11.12}$$

3rd
proof

and may be evaluated by means of Eqs. (11.11) and (11.12) above as a function of γ , p_e/p_0 , and p_a/p_0 . For the special case $\gamma = 1.2$, Fig. 11.3 shows how the diverging portion of the nozzle contributes to the rocket thrust at various area and pressure ratios. We can see that for each value of the ambient pressure ratio p_a/p_0 there is an optimum area ratio for the nozzle; it is the one for which $p_e = p_a$ (see discussion of Fig. 10.2).

We can see from Fig. 11.3 that the thrust of a rocket will actually be reduced by adding a diverging section of too large an area ratio. However for reasonable pressure ratios ($p_a/p_0 < 0.02$) this would not happen unless A_e/A^* were greater than about 30. For very small pressure ratios (such as $p_a/p_0 = 0.001$), we will see that very large area ratios are associated with maximum thrust.

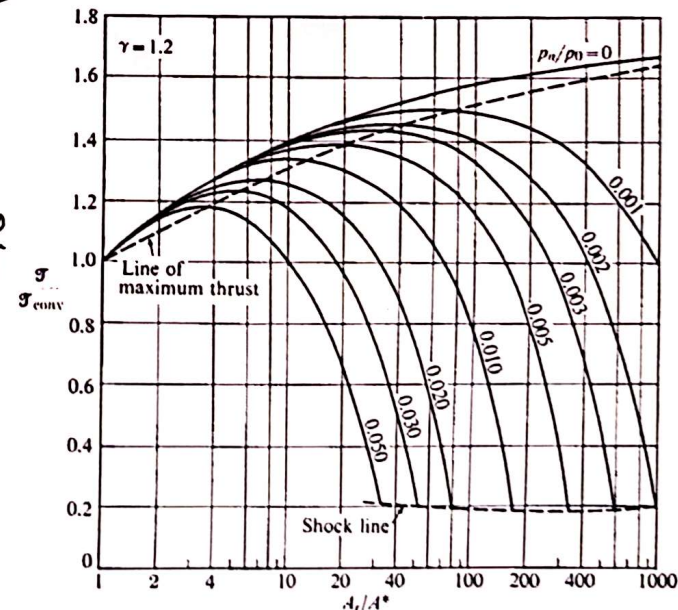
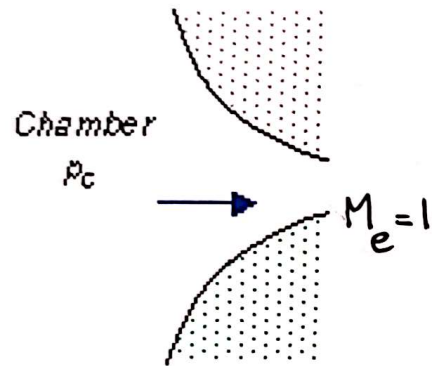
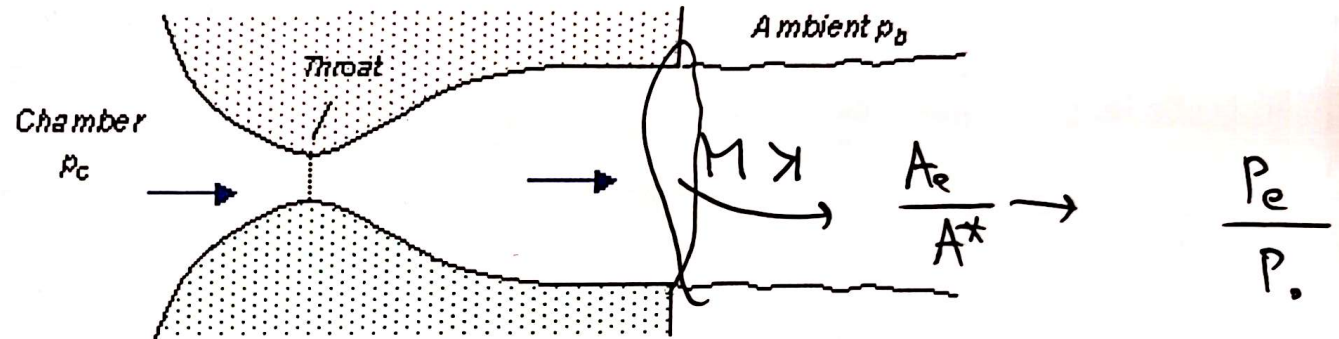


FIGURE 11.3 Performance characteristics of a one-dimensional isentropic rocket nozzle; $\gamma = 1.2$. (After Malina [1].)



Converging Nozzle



Converging-Diverging Nozzle

- Plot of thrust ratio vs. area ratio
 - Figure compares two non-dimensional numbers
 - Abscissa is ratio of nozzle exit area to minimum area, or nozzle exit area to throat area (minimum area always occurs at throat), A_e/A^*
 - Ordinate is ratio of thrust with diverging to converging nozzle, T/T_{conv}
- Curve is plotted for constant ratio of specific heats, $\gamma = c_p/c_v = 1.2$
 - Curve would shift for $\gamma = 1.4$ or any other value
- Curves correspond to various ratios of P_a/P_o
 - P_a/P_o = ambient (atmospheric) to combustion chamber pressure
 - P_o is approximately constant for most rockets

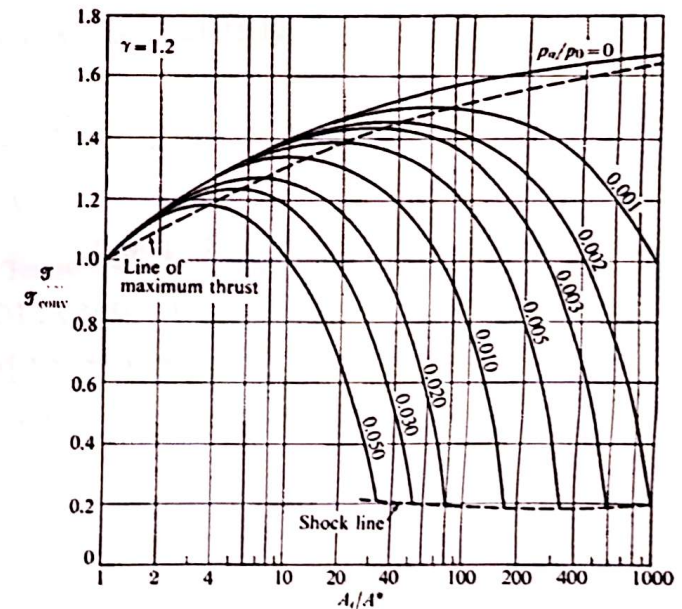


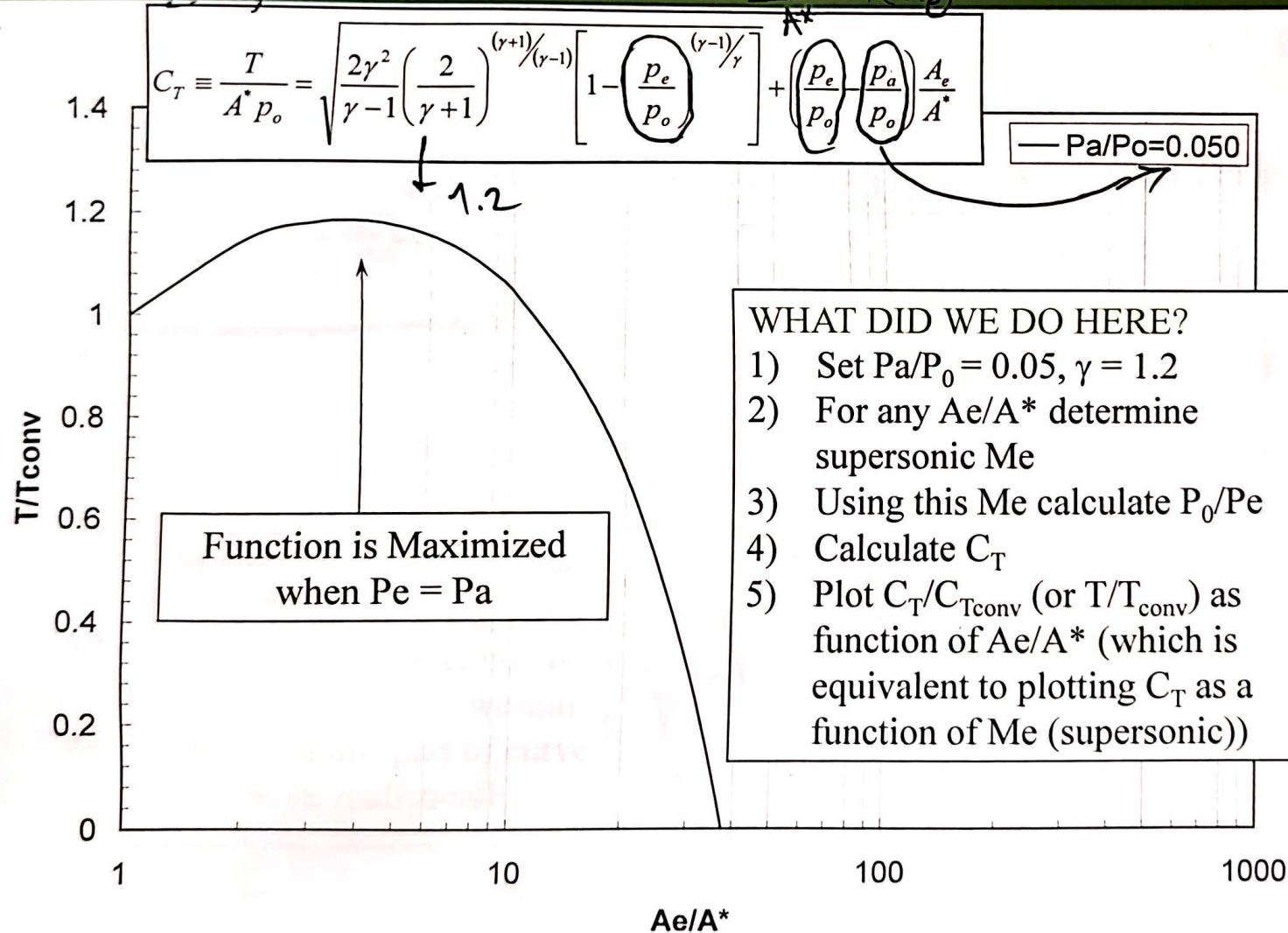
FIGURE 11.3 Performance characteristics of a one-dimensional isentropic rocket nozzle; $\gamma = 1.2$. (After Malina [1].)

$$\frac{P_e}{P_0} = \left(1 + \frac{\gamma-1}{2} M_e^2\right)^{\frac{\gamma}{\gamma-1}}$$

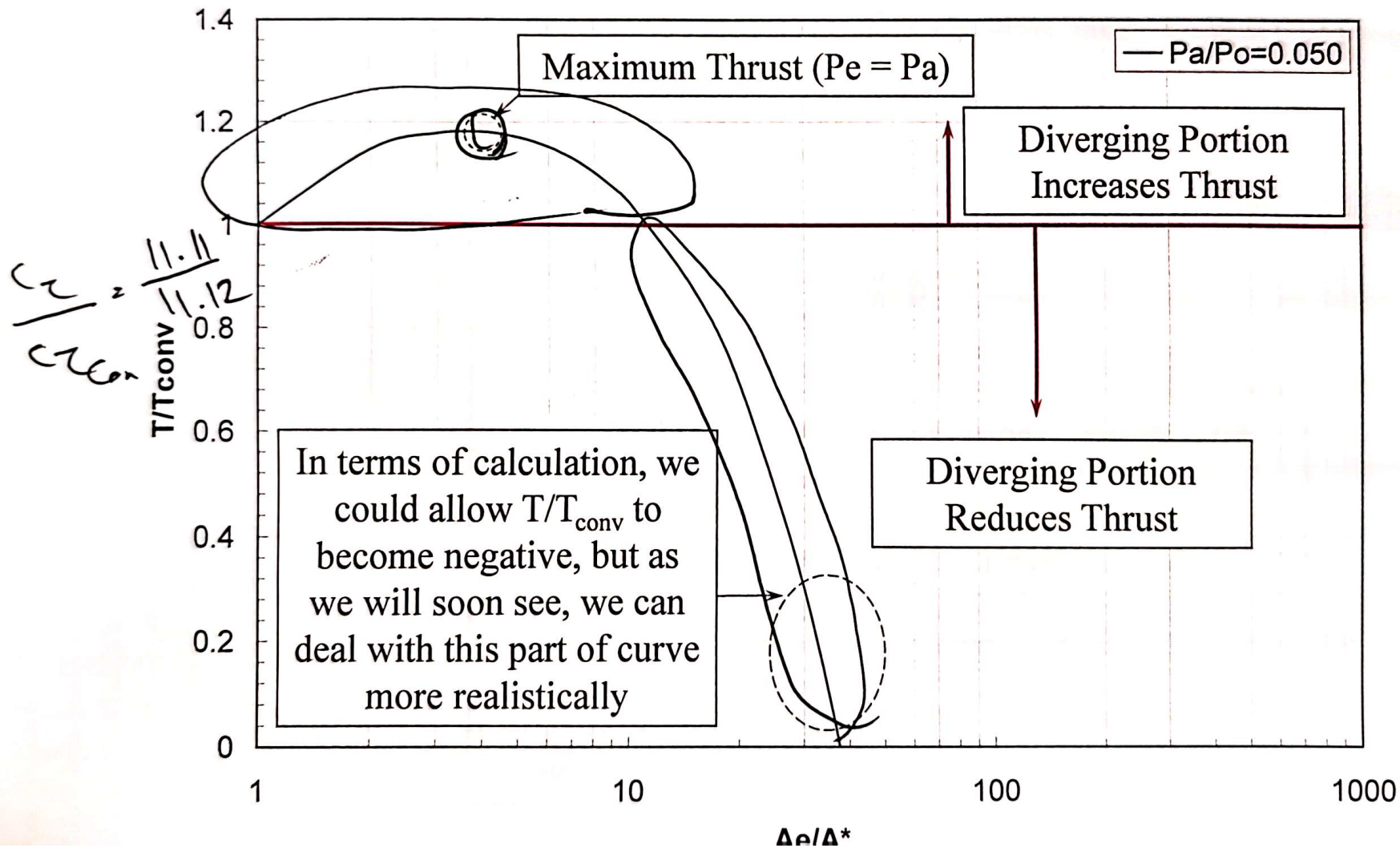
$$\frac{P_e}{P_0} \leftarrow \boxed{M_e} \leftarrow \frac{A_e}{A^*} \checkmark$$

$$\tau = \underline{m u_e} + \underline{(P_e - P_a) A_e}$$

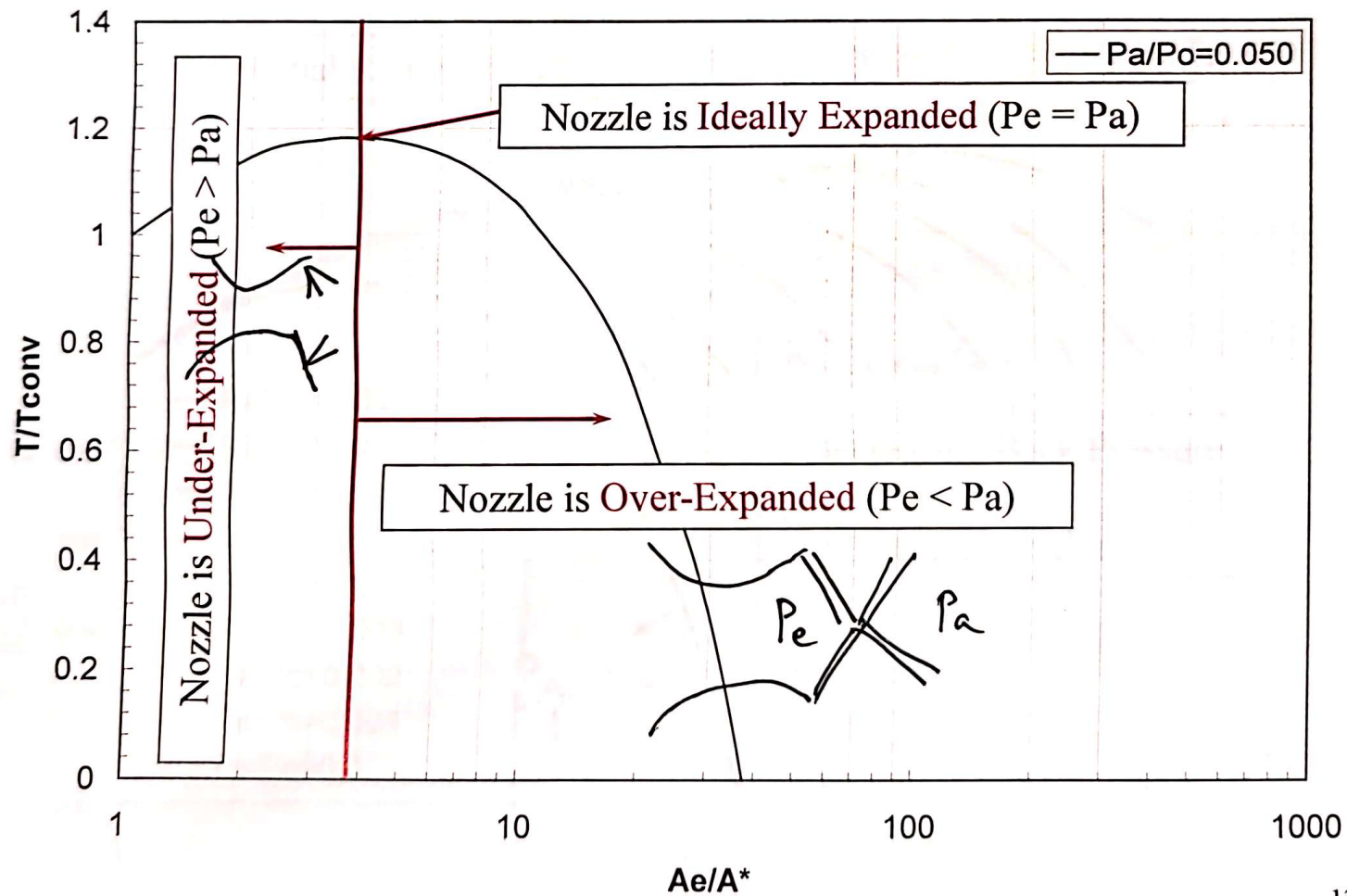
from $A_e = f(M_e)$ (sup.)



PERFORMANCE CHARACTERISTIC OF A 1-D ISENTROPIC NOZZLE



PERFORMANCE CHARACTERISTIC OF A 1-D ISENTROPIC NOZZLE



PERFORMANCE CHARACTERISTIC OF A 1-D ISENTROPIC NOZZLE

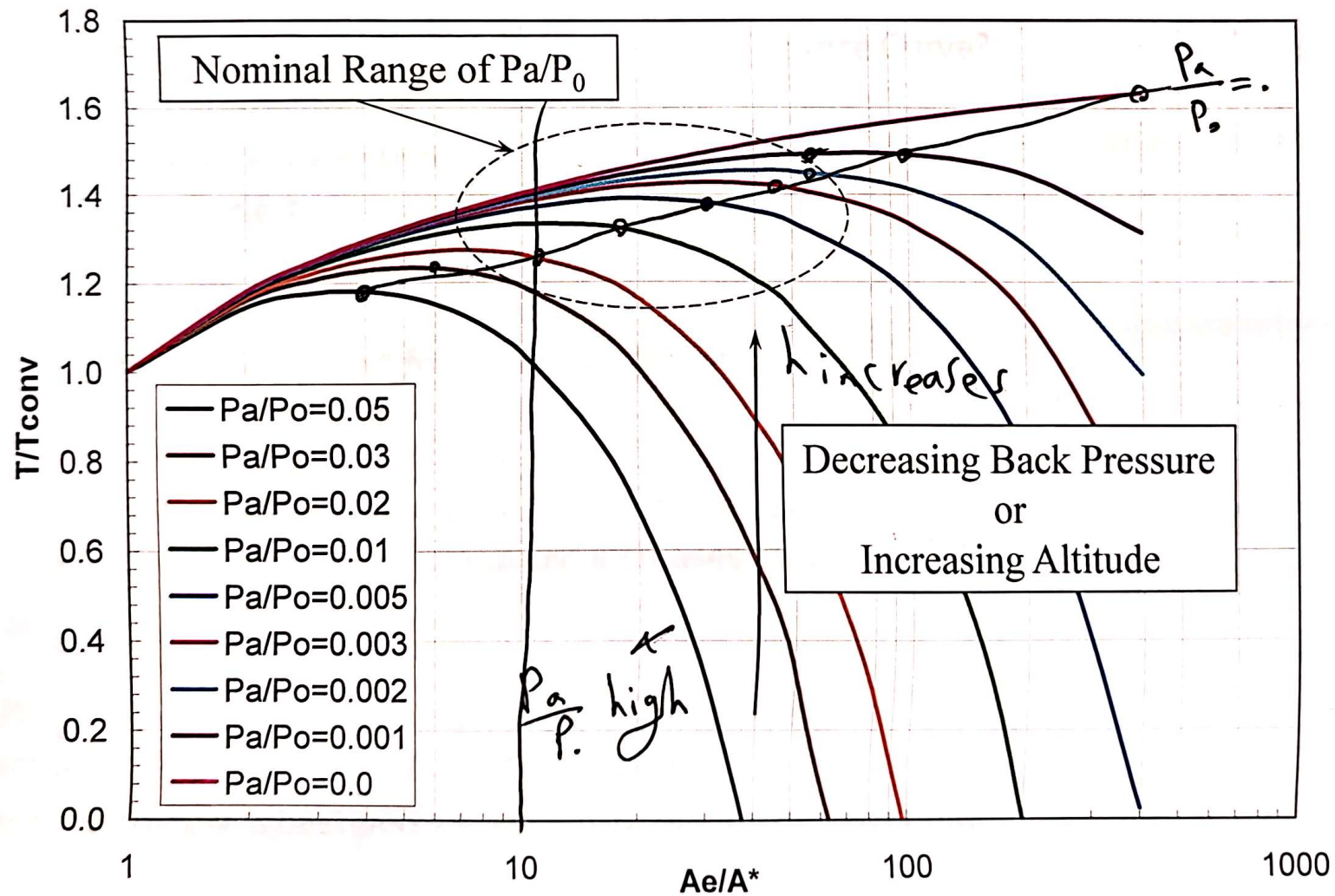


Figure 11.3 displays a shock line that shows the pressure ratios and area ratios for which a shock enters the nozzle. One may derive it as follows: Suppose that a normal shock sits in the exit plane of the nozzle and that the conditions just ahead of it are p_e , M_e , and so on; the pressure just behind it is p_a . Then, according to Eq. (3.36) for normal shocks,

$$\frac{A_e}{A^*} \rightarrow M_e > 1 \quad \left(\frac{p_a}{p_e} = \frac{2\gamma}{\gamma+1} M_e^2 - \frac{\gamma-1}{\gamma+1} \right)$$

For the isentropic nozzle the pressure ratio p_0/p_e is related to M_e by

$$\frac{p_0}{p_e} = \left(1 + \frac{\gamma-1}{2} M_e^2 \right)^{\gamma/(\gamma-1)}$$

so that $\frac{p_a}{p_e} \times \frac{p_e}{p_0}$

$$\frac{p_a}{p_0} = \left(\frac{2\gamma}{\gamma+1} M_e^2 - \frac{\gamma-1}{\gamma+1} \right) / \left(1 + \frac{\gamma-1}{2} M_e^2 \right)^{\gamma/(\gamma-1)} \quad (*)$$

M_e gives $\left\{ \frac{p_a}{p_0}, \frac{A_e}{A^*} \right\}$

Further, the area ratio is related to M_e by (applying Eq. 3.15)

$$\frac{A_e}{A^*} = \frac{1}{M_e} \left[\frac{2}{\gamma+1} \left(1 + \frac{\gamma-1}{2} M_e^2 \right) \right]^{(\gamma+1)/2(\gamma-1)}$$

These last two relations, then, show that for a given γ there is only one pressure ratio p_a/p_0 for which a normal shock will locate in the plane of a nozzle of a given area ratio A/A^* . Further discussion of back pressure effects follows in Section 11.3.

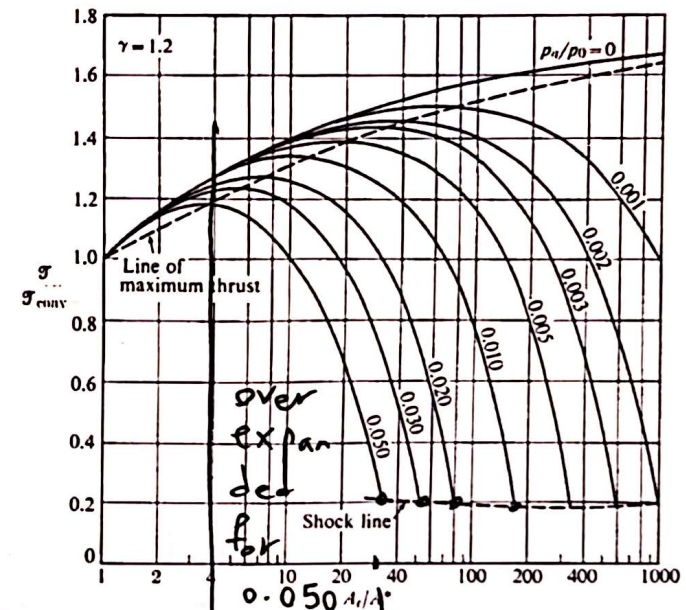
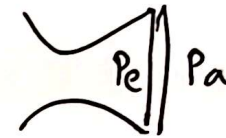


FIGURE 11.3 Performance characteristics of a one-dimensional isentropic rocket nozzle; $\gamma = 1.2$. (After Malina [1].)

Put $\frac{p_a}{p_0} = 0.05 \rightarrow$ get M_e from (*)

Put this M_e in Area-Mach relation, get $\frac{A_e}{A^*}$ that corresponds to shock at ex